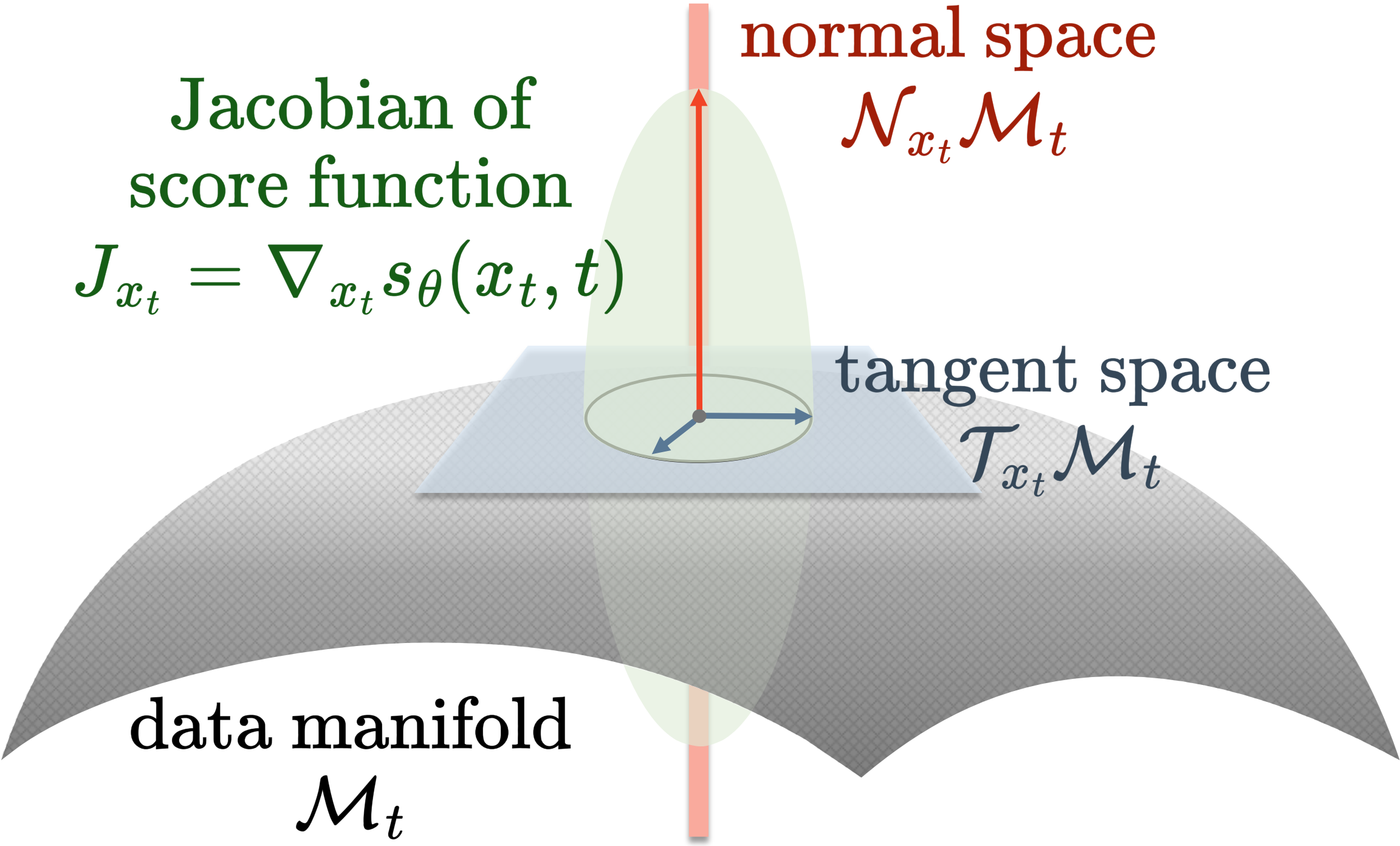
normal space
$\mathcal{N}_{x_t}\mathcal{M}_t$
Jacobian of
score function
$J_{x_t} = \nabla_{x_t} s_\theta(x_t, t)$
tangent space
$\mathcal{T}_{x_t}\mathcal{M}_t$
data manifold
$\mathcal{M}_t$